Chapter 3: Steady-State Equivalent Circuit Modeling, Losses, and Efficiency

Erickson & Maksimović — Fundamentals of Power Electronics

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Chapter Motivation

Chapter 3 builds a more powerful framework: the **equivalent circuit model**. The central idea is that any switching dc–dc converter can be represented at dc steady state as a simple circuit containing an *ideal dc transformer* plus lumped elements that model the various loss mechanisms. Once this model is in place, the full toolkit of conventional circuit analysis: Thevenin equivalents, voltage dividers, superposition, etc., becomes directly applicable.

3.1 The DC Transformer Model

Terminal Quantities of a Switching Converter

As illustrated in Fig. 3.1, any switching dc–dc converter has three ports:

- **Power input port**: voltage V_g, current I_g
- **Power output port**: voltage V, current I
- **Control input**: duty cycle D

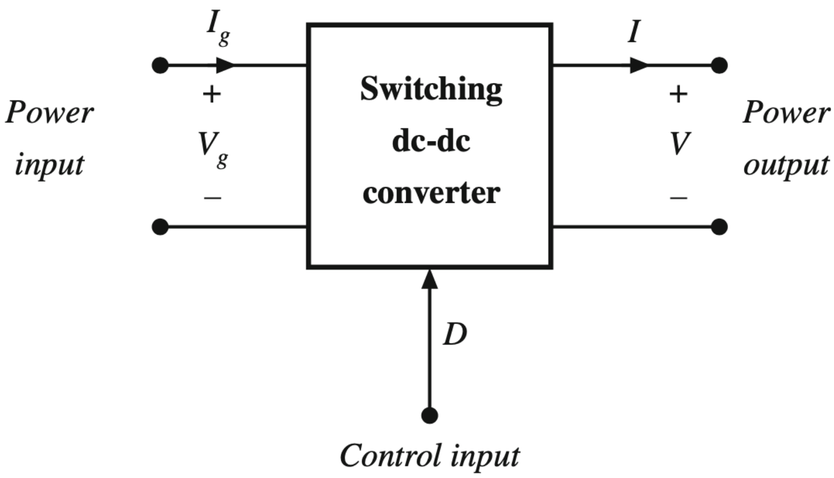

Figure 1: Switching converter terminal quantities

Under ideal (lossless) conditions, power conservation requires:

$$P_{\text{in}} = P_{\text{out}} \quad \Rightarrow \quad V_g I_g = VI \tag{1}$$

These relationships hold only under equilibrium (dc) conditions. During transients, the stored energy in inductors and capacitors changes, and these equations are temporarily violated.

Voltage and Current Conversion Ratios

The output voltage can always be written as:

$$V = M(D)\, V_g \tag{2}$$

where $M(D)$ is the **equilibrium conversion ratio**, a function only of the duty cycle D and independent of load for ideal converters operating in CCM. Substituting into the power balance equation yields:

$$I_g = M(D)\, I \tag{3}$$

Converter	$M(D)$	Condition
Buck	D	$M \leq 1$
Boost	$1/(1-D)$	$M \geq 1$
Buck-Boost	$-D/(1-D)$	Sign-inverted

Identification with the Ideal Transformer

Equations are *identical* in form to the terminal equations of an ideal transformer with turns ratio $1 : M(D)$.

Although a physical transformer cannot pass dc (its core saturates), we are free to define an idealised model element: the DC Transformer, that obeys the same port equations for dc quantities.

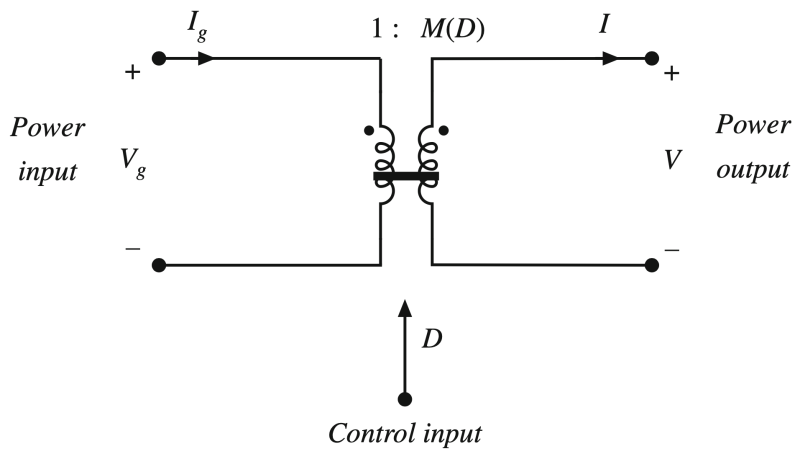

Figure 2: Ideal dc transformer model of a switching converter

DC Transformer Model (ideal): any switching dc–dc converter operating in steady-state CCM behaves at its terminals exactly like an ideal transformer with turns ratio $1 : M(D)$, where M depends only on D and is independent of load.

Using the Model in a Network

The power of this model is that the dc transformer can be inserted into an external circuit and all standard transformer manipulation rules apply directly.

Example: A Thevenin source V_1, R_1 drives the converter, which feeds a resistive load R.

1. Replace the switching converter with the dc transformer symbol (turns ratio $1 : M(D)$).
2. Refer all primary-side elements to the secondary: $V_1 \to M(D)V_1$, $R_1 \to M^2(D)R_1$.
3. Solve the resulting simple resistive divider:

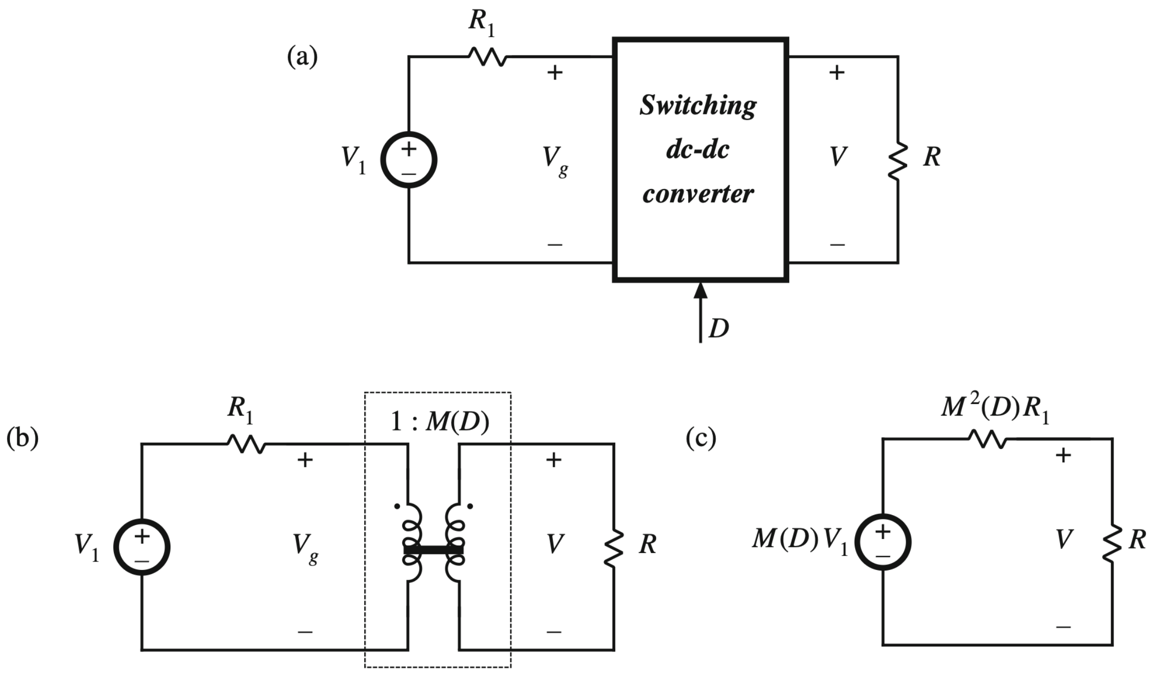

Figure 3: (a) Original circuit with switching converter; (b) substitution of DC transformer model; (c) simplification by referring all elements to the secondary side

$$V = M(D)\, V_1 \cdot \frac{R}{R + M^2(D)\, R_1} \tag{4}$$

Key insight: the dc transformer model converts a non-linear switching circuit into a *linear, time-invariant* equivalent. All network theorems (Thevenin, Norton, superposition, voltage divider) apply directly to the equivalent circuit.

3.2 Inclusion of Inductor Copper Loss

Physical Origin of Copper Loss

A real inductor exhibits two categories of power loss:

1. **Copper loss** — resistive dissipation in the winding wire, characterised by the dc resistance R_L (also called the inductor winding resistance or DCR).
2. **Core loss** — hysteresis and eddy-current losses in the magnetic core material.

   The copper loss is modeled by placing a resistor R_L in series with the ideal inductor L:

Analysis of the Non-Ideal Boost Converter

The inductor model of 4 is inserted into the Boost converter circuit. The analysis proceeds identically to Chapter 2, using volt-second balance and charge balance, but now with R_L appearing in the inductor voltage expressions.

Figure 4: Inductor copper loss model: ideal inductor L in series with winding resistance R_L

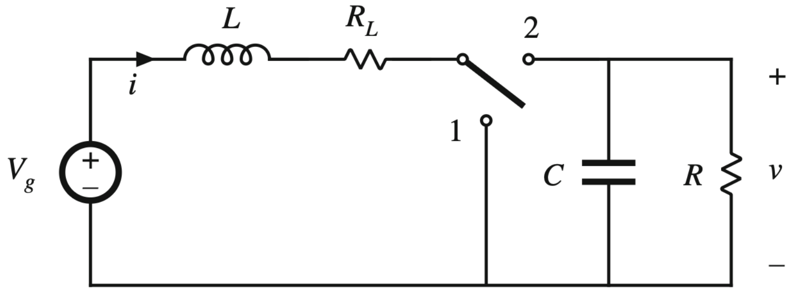

Figure 5: Boost converter with inductor winding resistance R_L

Sub-interval 1: $0 < t < DT_s$ (switch in position 1)

The diode is reverse-biased; the inductor is connected directly to V_g. Applying the small-ripple approximation ($i(t) \approx I$, $v(t) \approx V$):

$$v_L(t) = V_g - i(t)\,R_L \approx V_g - IR_L \tag{5}$$

$$i_C(t) = -\frac{v(t)}{R} \approx -\frac{V}{R} \tag{6}$$

Sub-interval 2: $DT_s < t < T_s$ (switch in position 2)

The diode conducts; the inductor is connected to the output.

$$v_L(t) \approx V_g - IR_L - V \tag{7}$$

$$i_C(t) \approx I - \frac{V}{R} \tag{8}$$

Averaging the inductor voltage over one period and setting to zero:

$$\langle v_L \rangle = \frac{1}{T_s}\int_0^{T_s} v_L\,dt = D(V_g - IR_L) + D'(V_g - IR_L - V) = 0 \tag{9}$$

Expanding and collecting terms (noting $D + D' = 1$):

$$\boxed{0 = V_g - IR_L - D'V} \tag{10}$$

Compared to the ideal Boost result ($0 = V_g - D'V$), the copper loss introduces the additional term $-IR_L$, which reduces the achievable output voltage.

$$\langle i_C \rangle = D\left(-\frac{V}{R}\right) + D'\left(I - \frac{V}{R}\right) = 0 \tag{11}$$

Collecting terms:

$$\boxed{0 = D'I - \frac{V}{R}} \tag{12}$$

Eliminating I, giving $I = V/(D'R)$ and substituting:

$$\boxed{\frac{V}{V_g} = \frac{1}{D'} \cdot \frac{1}{1 + \frac{R_L}{D'^2 R}}} \tag{13}$$

The voltage conversion ratio is the ideal value $1/D'$ multiplied by a *loss correction factor* that is always ≤ 1. The factor approaches 1 when $R_L \ll D'^2 R$ and degrades significantly when R_L is comparable to $D'^2 R$.

Behaviour at Extremes of Duty Cycle

Limit	V/V_g	Physical explanation
$R_L \to 0$	$\to 1/D'$	Ideal Boost result recovered
$D \to 0$	$\to 1$	Switch always open; $V \approx V_g$
$D \to 1$	$\to 0$	Switch always closed; energy never transferred to output; all power dissipated in R_L

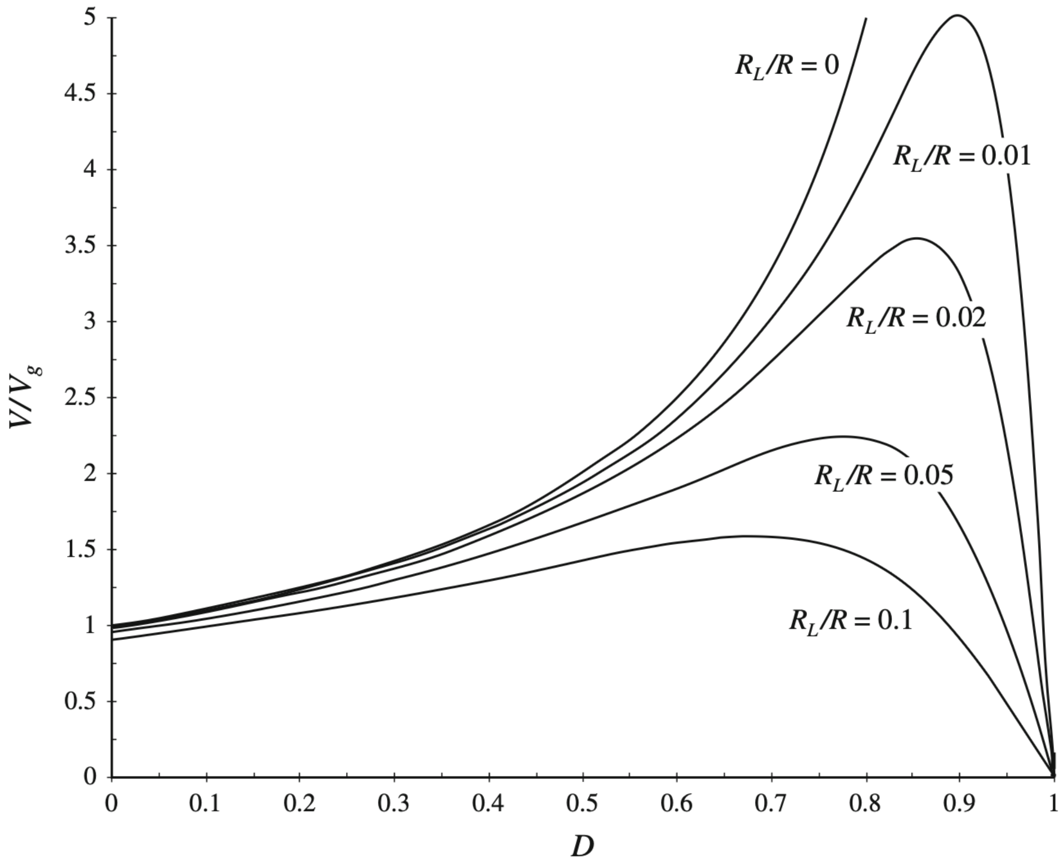

Figure 6: Output voltage ratio V/V_g vs. duty cycle D

The curve for $R_L = 0$ diverges as $D \to 1$, while every curve with $R_L > 0$ rises, reaches a peak, and then falls back to zero. This is the fundamental practical

limitation of the Boost converter: copper loss imposes a ceiling on the achievable conversion ratio.

The maximum of V/V_g occurs at a duty cycle $D^* < 1$, and the maximum conversion ratio is:

$$\left.\frac{V}{V_g}\right|_{\max} \approx \frac{1}{2}\sqrt{\frac{R}{R_L}} \quad \text{(approximately, for small } R_L/R\text{)}$$

For example, with $R_L/R = 0.02$, the maximum ratio is approximately 3.5. To achieve $V/V_g = 5$, one requires $R_L < 0.01R$, necessitating a larger, heavier, more expensive inductor. This is the classic performance–cost–size tradeoff in converter design.

3.3 Construction of the Equivalent Circuit Model

3.3.1 Inductor Voltage Equation → Voltage Loop

Restate (10):

$$\langle v_L \rangle = 0 = V_g - IR_L - D'V \tag{14}$$

This is a KVL equation around a loop with dc loop current I. Identify each term:

Term	Sign	Circuit element
V_g	$+$	Independent voltage source
IR_L	$-$	Resistor R_L carrying current I
$D'V$	$-$	Dependent voltage source (proportional to output voltage V)

3.3.2 Capacitor Current Equation → Current Node

Restate (12):

$$\langle i_C \rangle = 0 = D'I - \frac{V}{R} \tag{3.17}$$

This is a KCL equation at a node with dc node voltage V. Identify each term:

Term	Direction	Circuit element
$D'I$	into node	Dependent current source (proportional to inductor current I)
V/R	out of node	Load resistor R

3.3.3 Complete Circuit Model

Combining KVL and KCL into a single circuit 7 yields a circuit containing two dependent sources:

- A dependent *voltage* source of value $D'V$ (voltage-controlled, coefficient D')
- A dependent *current* source of value $D'I$ (current-controlled, coefficient D')

The equivalent circuit can be solved by referring all primary-side elements to the secondary side of the $D' : 1$ transformer:

- $V_g \to V_g/D'$ (voltage source divided by turns ratio)
- $R_L \to R_L/D'^2$ (resistance divided by turns ratio squared)

The inductor current I can be found by referring all elements to the primary side instead:

$$I = \frac{V_g}{D'^2 R + R_L} \cdot \frac{1}{1 + \dfrac{R_L}{D'^2 R}} \tag{15}$$

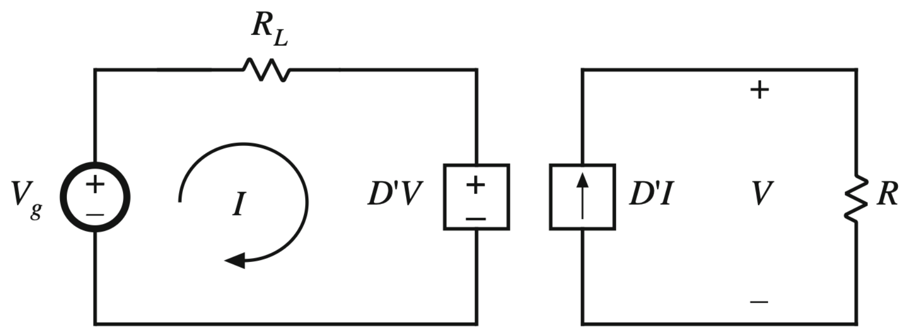

Figure 7: The two sub-circuits combined into a single equivalent circuit

Advantage of the circuit model: no algebra is needed beyond a voltage divider. Furthermore, the circuit is reusable adding more loss elements simply means inserting more resistors or voltage sources into the existing model, without rebuilding the analysis from scratch.

3.3.4 Efficiency

The equivalent circuit model allows efficiency to be computed directly by inspection:

$$P_{\text{in}} = V_g \cdot I \tag{16}$$
$$P_{\text{out}} = V \cdot (D'I) \tag{17}$$

The load current equals $D'I$ because it is drawn from the secondary of the $D' : 1$ transformer. Therefore:

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V \cdot D'I}{V_g \cdot I} = \frac{V}{V_g} \cdot D' \tag{18}$$

Substituting:

$$\boxed{\eta = \frac{1}{1 + \dfrac{R_L}{D'^2 R}}} \tag{19}$$

Interpretation: efficiency depends only on the ratio $R_L/(D'^2R)$. To achieve high efficiency, the inductor winding resistance must be much smaller than the *effective load resistance referred to the primary*, D'^2R. Note that as $D \to 1$, $D' \to 0$ and $D'^2R \to 0$, so the denominator grows and efficiency collapses to zero, consistent with the physical picture.

3.4 How to Obtain the Input Port of the Model

For the Boost converter, the input port of the dc transformer emerged automatically because the input current $i_g(t)$ coincides with the inductor current $i(t)$. This is not always the case.

For the Buck converter, the input current $i_g(t)$ is *not* a continuous waveform. The dc component of $i_g(t)$ must be computed explicitly from the input current waveform before the primary of the dc transformer model can be identified.

Buck Converter with Copper Loss

Volt-Second Balance (Inductor)

Sub-interval 1 ($0 < t < DT_s$, switch closed):

$$v_L = V_g - I_L R_L - V_C$$

Sub-interval 2 ($DT_s < t < T_s$, switch open):

$$v_L = -I_L R_L - V_C$$

Volt-second balance ($\langle v_L \rangle = 0$):

$$0 = DV_g - I_L R_L - V_C \tag{20}$$

Charge Balance (Capacitor)

$$0 = I_L - \frac{V_C}{R} \tag{21}$$

Finding the Input Port from the Input Current Waveform

The dc component of the input current $i_g(t)$ is found from its waveform:

$$I_g = \frac{1}{T_s} \int_0^{T_s} i_g(t)\, dt = \frac{DT_s \cdot I_L}{T_s} = DI_L \tag{22}$$

This gives a dependent current source $I_g = DI_L$ at the input port. Combined with the dependent voltage source DV_g, the pair is identified as the primary of a $1 : D$ dc transformer.

Combining the input port circuit with the output port circuit yields, which simplifies to the complete model of 8 with a $1 : D$ ideal dc transformer:

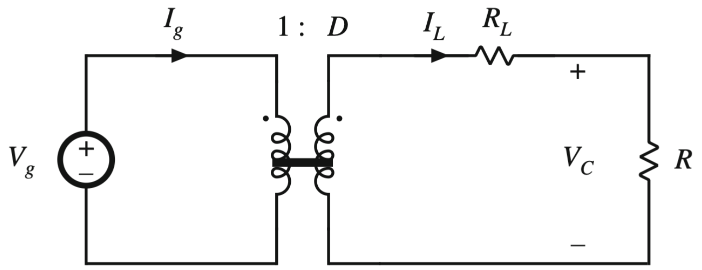

Figure 8: Complete equivalent circuit model of the Buck converter

Structural difference between Buck and Boost models:

	Buck	Boost
Transformer ratio	$1 : D$	$D' : 1$
R_L location	**secondary** side	**primary** side
Reason	Inductor is in output branch	Inductor is in input branch

The location of R_L in the model is determined solely by the physical position of the inductor in the converter circuit. No arbitrary choice is made.

Solution for Buck Output Voltage

Referring all elements to the secondary side of the $1 : D$ transformer (the transformer has no primary-side resistors in this case):

$$V_C = DV_g \cdot \frac{R}{R + R_L} = \frac{DV_g}{1 + R_L/R} \tag{23}$$

Note the absence of a D^2 factor in the denominator, compared to the Boost result. This is because R_L is already on the secondary and requires no referral transformation.

For the Buck converter, efficiency is similarly given by:

$$\eta_{\text{Buck}} = \frac{R}{R + R_L} = \frac{1}{1 + R_L/R}$$

3.5 Inclusion of Semiconductor Conduction Losses

Sources of Semiconductor Loss

In addition to inductor copper loss, practical converters experience significant power dissipation in semiconductor devices. Two mechanisms dominate in steady-state (conduction) operation:

Device	Loss mechanism	Model	Circuit element
MOSFET, BJT	On-resistance	$P = I^2 R_{on}$	Series resistor R_{on}
Diode, IGBT	Forward voltage drop + resistance	$P = V_D I + I^2 R_D$	Series V_D + R_D

Semiconductor switching losses exist but are not modelled here; they are treated in later chapters on magnetics and converter design.

Boost Converter with All Conduction Losses

The Boost converter with MOSFET on-resistance R_{on}, diode forward voltage V_D, and diode resistance R_D is shown in 9.

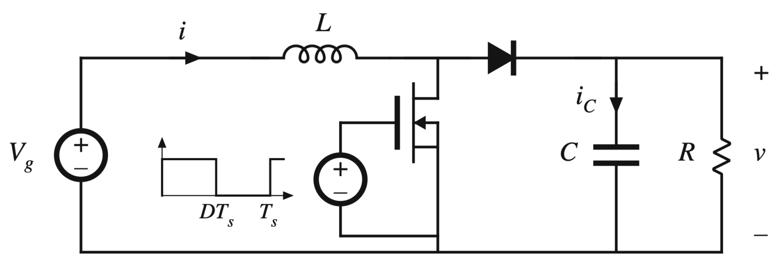

Figure 9: Boost converter including MOSFET on-resistance R_{on} and diode model $V_D + R_D$

Sub-interval 1: MOSFET ON, Diode OFF $(0 < t < DT_s)$

The MOSFET is in the conduction state, modelled by R_{on}. The diode is reverse-biased.

$$v_L(t) = V_g - iR_L - iR_{on} \approx V_g - IR_L - IR_{on} \tag{24}$$

$$i_C(t) \approx -\frac{V}{R} \tag{25}$$

Sub-interval 2: MOSFET OFF, Diode ON ($DT_s < t < T_s$)

The MOSFET turns off; the inductor forces the diode into conduction, modelled by $V_D + R_D$.

$$v_L(t) = V_g - iR_L - V_D - iR_D - v \approx V_g - IR_L - V_D - IR_D - V \tag{26}$$

$$i_C(t) \approx I - \frac{V}{R} \tag{27}$$

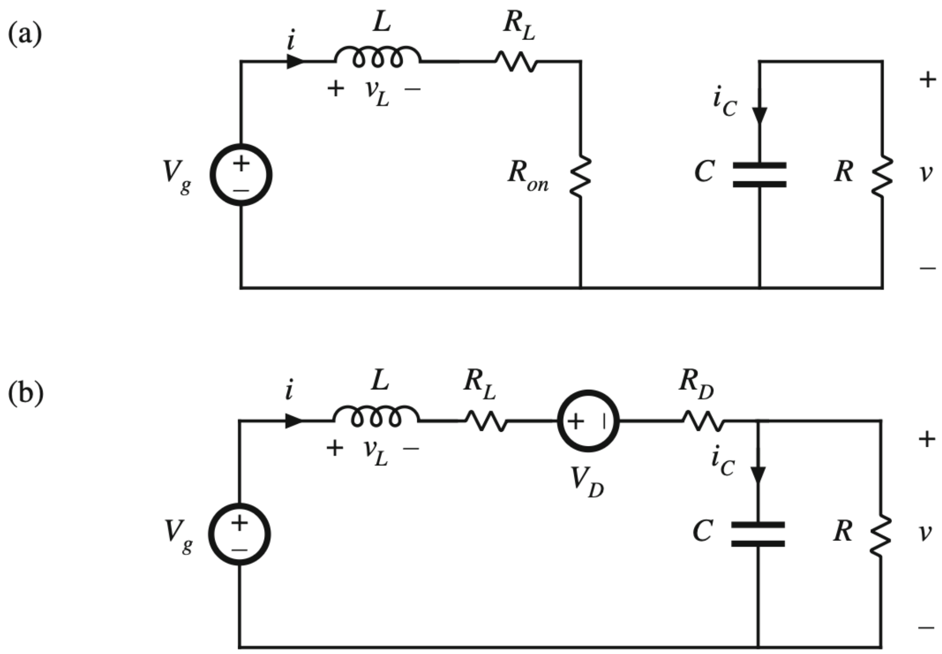

Figure 10: Boost converter sub-interval circuits: (a) MOSFET conducting; (b) diode conducting

Volt-Second Balance with All Losses

$$\langle v_L \rangle = D(V_g - IR_L - IR_{on}) + D'(V_g - IR_L - V_D - IR_D - V) = 0 \tag{28}$$

Expanding and collecting (using $D + D' = 1$):

$$\boxed{V_g - IR_L - IDR_{on} - D'V_D - ID'R_D - D'V = 0} \tag{29}$$

Charge Balance

The capacitor charge balance is identical in form to Section 3.3 (the capacitor circuit is unchanged by the semiconductor models):

$$D'\,I - \frac{V}{R} = 0 \tag{30}$$

Following the same procedure as Section 3.3. The resulting equivalent circuits are shown in 11.

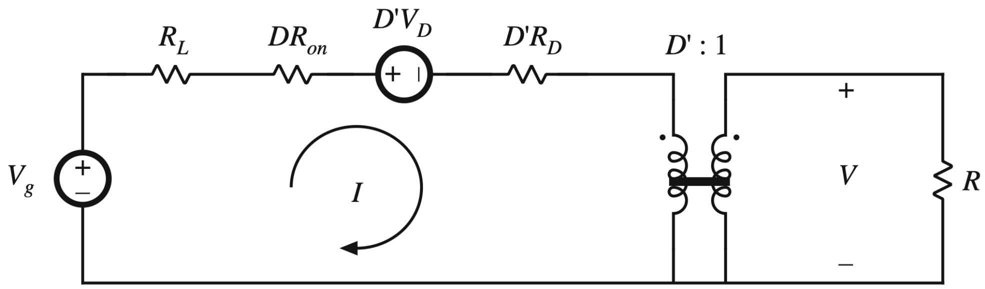

Figure 11: Complete equivalent circuit model of the Boost converter

Physical Interpretation of Effective Resistances

The MOSFET conducts only during sub-interval 1 (fraction D of the period). Its average power loss is $I^2 \cdot DR_{on}$, equivalent to a resistance DR_{on} carrying dc current I for the full period.

The diode conducts only during sub-interval 2 (fraction D'). Its average conduction loss is $I^2 \cdot D'R_D$, equivalent to $D'R_D$ in the dc model.

Output Voltage and Efficiency

Solving 11 by referring all elements to the secondary side (dividing voltages by D' and resistances by D'^2):

$$V = \frac{1}{D'}\left(V_g - D'V_D\right) \cdot \frac{D'^2 R}{D'^2 R + R_L + DR_{on} + D'R_D} \tag{31}$$

Dividing by V_g:

$$\boxed{\frac{V}{V_g} = \frac{1}{D'}\left(1 - \frac{D'V_D}{V_g}\right) \cdot \frac{1}{1 + \dfrac{R_L + DR_{on} + D'R_D}{D'^2 R}}} \tag{32}$$

The conversion ratio is the ideal value $1/D'$, multiplied by two correction factors:

1. $\left(1 - \dfrac{D'V_D}{V_g}\right)$: reduction due to diode voltage drop V_D.
2. $\dfrac{1}{1 + (R_L + DR_{on} + D'R_D)/(D'^2 R)}$: reduction due to all resistive losses.

All loss elements act to reduce V/V_g below the ideal value $1/D'$.

Efficiency:

$$\eta = D'\frac{V}{V_g} = \frac{\left(1 - \dfrac{D'V_D}{V_g}\right)}{1 + \dfrac{R_L + DR_{on} + D'R_D}{D'^2 R}} \tag{33}$$

For high efficiency, both conditions must hold simultaneously:

$$\frac{V_g}{D'} \gg V_D \qquad \text{and} \qquad D'^2 R \gg R_L + DR_{on} + D'R_D \tag{34}$$

Accuracy of the Average Model: RMS vs. Average Currents

A subtle issue arises because power dissipation in a resistor is $P = I_{\text{rms}}^2 R$, yet the equivalent circuit uses only the dc average current I. The model predicts $P = I^2 \cdot DR_{on}$ for the MOSFET, while the true loss is $I_{\text{rms}}^2 R_{on}$.

Case	Inductor ripple Δi	MOSFET I_{rms}	Actual loss in R_{on}
(a)	$\Delta i = 0$ (infinite L)	$I\sqrt{D}$	$DI^2 R_{on}$ (exact match)
(b)	$\Delta i = 0.1\,I$	$1.00167\,I\sqrt{D}$	$1.0033 \cdot DI^2 R_{on}$ (+0.33%)
(c)	$\Delta i = I$	$1.155\,I\sqrt{D}$	$1.333 \cdot DI^2 R_{on}$ (+33%)

For small current ripple (the small-ripple approximation), the average model predicts losses accurately to within a fraction of a percent. The model is valid as long as the inductor is sized to keep the ripple small, which is already a design requirement for good converter performance.

3.6 Summary of Key Principles

1. **DC Transformer Model:** in steady-state CCM, any switching converter is equivalent to an ideal dc transformer with turns ratio $1 : M(D)$. The conversion ratio M depends only on D and is independent of load. This model enables direct use of circuit analysis techniques.

2. **Modelling non-idealities:** every source of power loss or dynamic behaviour is incorporated by *adding a physical circuit element* to the ideal model. Resistors model conduction losses; voltage sources model semiconductor forward drops; inductors and capacitors model dynamics. The structure of the model is preserved.

3. **Volt-second balance $\leftrightarrow$ KVL; charge balance $\leftrightarrow$ KCL:** the steady-state balance equations are KVL/KCL equations in disguise. They can be directly translated into loop and node circuits, which are then combined into the complete equivalent circuit.

4. **Effective resistances:** semiconductor on-resistances appear multiplied by the duty cycle of the sub-interval during which they conduct. These effective values correctly represent the time-averaged power dissipation under the small-ripple approximation.

5. **Efficiency and accuracy:** the average equivalent circuit model accurately predicts efficiency and output voltage provided the inductor current ripple is small. Errors introduced by using average rather than rms currents are negligible for well-designed converters (typically $< 1\%$ for $\Delta i < 0.1I$).

6. **Loss hierarchy in a Boost converter (all mechanisms combined):**

$$\frac{V}{V_g} = \underbrace{\frac{1}{D'}}_{\text{ideal}} \times \underbrace{\left(1 - \frac{D'V_D}{V_g}\right)}_{\text{diode drop}} \times \underbrace{\frac{1}{1 + (R_L + DR_{on} + D'R_D)/(D'^2 R)}}_{\text{resistive losses}}$$